

# Decision Log: Aerial GCP Pose Estimation

## Candidate Information

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## 1. Problem Understanding

The assignment required solving two tasks simultaneously:

1. Predict the pixel coordinates (x, y) of the Ground Control Point (GCP) marker center.
2. Classify the marker shape into one of three classes:
  - Cross
  - Square
  - L-Shape

This was formulated as a **multi-task learning problem** involving both regression and classification.

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## 2. Dataset Analysis

An Exploratory Data Analysis (EDA) was performed before model development.

### Key Findings

- Total annotations: **1000**
- Valid samples used: **996**
- Missing labels: **4**
- Test images: **300**

### Shape Distribution

| Shape | Count |
|-------|-------|
|-------|-------|

|       |     |
|-------|-----|
| Cross | 177 |
|-------|-----|

|        |     |
|--------|-----|
| Square | 328 |
|--------|-----|

## Shape Count

L-Shape 491

The dataset exhibited class imbalance, particularly for the **Cross** class.

Two major image resolutions were observed:

- 4096 × 2730
  - 4096 × 3068
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### 3. Handling Missing Labels

Four samples lacked verified\_shape labels.

Decision:

- Excluded these samples from training.
- Retained only fully annotated data.

Reason:

Training classification models with missing labels could introduce noise and degrade performance.

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### 4. Model Selection

The final model uses **EfficientNet-B0 pretrained on ImageNet** as the backbone.

Reasons for selection:

- Lightweight and computationally efficient
- Strong feature extraction capabilities
- Suitable for drone imagery
- Faster training and inference compared to larger architectures

The model was designed as a **multi-task network** with:

- Regression head for (x, y) prediction
  - Classification head for marker shape prediction
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### 5. Data Augmentation Strategy

The following augmentations were applied during training:

- Horizontal Flip
- Vertical Flip
- Rotation
- Brightness/Contrast Adjustment
- Gaussian Blur
- Resize to  $512 \times 512$

These augmentations improve model robustness against viewpoint and lighting variations in drone imagery.

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## 6. Handling Class Imbalance

Class imbalance was addressed using **weighted cross-entropy loss**.

Computed class weights:

- Cross: 1.8685
- Square: 1.0127
- L-Shape: 0.6769

This ensures fair learning across all classes.

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## 7. Loss Function Design

The model optimizes a weighted multi-task loss:

Total Loss =

$\text{REG\_WEIGHT} \times \text{Regression Loss} + \text{CLS\_WEIGHT} \times \text{Classification Loss}$

Regression Loss:

- Smooth L1 Loss

Classification Loss:

- Weighted Cross Entropy Loss
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## 8. Hyperparameter Experiments

Multiple loss weight configurations were evaluated:

### Experiment 1

REG\_WEIGHT = 1

CLS\_WEIGHT = 1

### Experiment 2

REG\_WEIGHT = 5

CLS\_WEIGHT = 1

### Experiment 3

REG\_WEIGHT = 10

CLS\_WEIGHT = 1

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## 9. Final Model Selection

The final model selected:

REG\_WEIGHT = 5

CLS\_WEIGHT = 1

Reasons:

- Achieved the lowest validation loss
- Maintained perfect classification performance
- Provided better balance between localization and classification tasks

Final performance:

- Best Validation Loss: **0.1729**
- Macro F1 Score: **1.0000**

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## 10. Evaluation Metrics

Classification Metrics:

- Accuracy
- Macro F1 Score

Localization Metrics:

- PCK@10
- PCK@25
- PCK@50

These metrics were chosen to evaluate both classification accuracy and localization quality.

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## 11. Inference Pipeline

The trained model was used to generate predictions for all **300 test images**.

The output was saved in predictions.json, following the exact schema specified in the assignment.

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## 12. Conclusion

A robust multi-task deep learning pipeline was developed for GCP localization and shape classification. The final system successfully fulfills all assignment requirements and demonstrates strong performance on both tasks.